AvatarX Cardiac Platform — Technical Implementation Plan

Audience: Engineering team · **Date:** August 15, 2026 · **Status:** v1.0 for team review **Repo:** `AvatarX/code/afib` (extend in place — do not fork) · **Companion docs:** full feasibility review + plain-English brief (`AvatarX/research/`), execution prompts v0.2 / v0.2.1 / v0.3 (repo root)

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1. What we are building, in one paragraph

A contactless cardiac platform: **30–60 s smartphone face scan → validated rhythm findings**, with AFib screening as the flagship claim, built on an architecture that (a) collects facial video paired with reference ECG at clinical quality, (b) retrains and promotes models only when they beat pre-registered baselines on unseen patients, and (c) contains an ECG-reconstruction research track whose synthetic output is **promotion-gated** — it can reach a user surface only if it passes clinical-fidelity gates on held-out patients. The engineering north star is a single principle every module enforces: **the system renders what it measures, infers only what the measured signal supports, and abstains (NO_RESULT) rather than guesses.** This is not compliance decoration; it is the product strategy (no company holds an FDA clearance for contactless rhythm detection — the lane is open for whoever shows up with defensible evidence).

2. System architecture

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L0  Capture & Sessions      capture/          EXISTS   frame ingest, face tracking, PRBS/LED sync
L1  Signal extraction       rppg/             EXISTS   POS/CHROM on tracked multi-ROI
                            rbcg/             STUB     ablation-only (patent review pending)
L2  Quality & gates         preprocessing/,   EXISTS   fail-closed capture/sync/SQI gates
                            inference/                 + fairness telemetry (NEW)
L3  Beat lattice            beats/            EXISTS   beat detection, clean runs, confidence
L4  Endpoint heads          heads/            NEW      registry: afib | rate_flags | findings
                                                       (+ flutter_suspicion, burden — research)
L5  Decision & report       inference/, app/  EXISTS   ScanResult v2 (NEW), findings-only report (NEW)
L6  Data engine             datasets/         PARTIAL  campaigns, paired-ECG ingest, registry (NEW)
L7  Training engine         training/         NEW      runs, baselines, promotion, model registry
L8  Reconstruction lab      research/         NEW      video→ECG decoder + §G gates (quarantined)
```

Data flow: `video + manifest → L0 → L1 → L2 (ACCEPT / REPEAT_SCAN / NO_RESULT) → L3 BeatLattice → registered L4 heads → L5 ScanResult v2 → findings report`. L6/L7 run offline; L8 consumes L1–L3 outputs and paired ECG, and writes only to `research/runs/`.

Current state: v0.1.5 core is real and green — 139 tests; synchronized capture with PRBS LED markers; POS/CHROM rPPG; anti-periodicity SQI; beat detection with confidence calibration; Model A (trained on degraded real RR data, MIMIC PERform AF); decision logic with sanctioned sentences and 1–5 star confidence; live browser demo; evaluation harness (Gate-1/1b, risk-coverage, no-read parity, leakage audit). All published numbers so far derive from synthetic data or the E6 public-RR experiment — no clinical performance is claimed.

3. Standing invariants (every workstream; each is or becomes a test)

1. Never emit a diagnosis; all user-facing strings flow through `ScanResult.user_facing_text()`.
2. Fail closed: invalid capture/sync, <5 clean intervals, or NaN → `NO_RESULT` with reasons.
3. The gated production path is the only path metrics are reported from.
4. Features run on clean runs only; no interval repair (in AF, the "outlier" interval *is* the signal).
5. SQI must never score periodicity (AF is aperiodic; a periodicity-based quality score screens out the patients we exist to find).
6. Participant-level (and session-level) splits everywhere, including quick experiments.

7. Measurement-class labeling on every output field: `MEASURED` / `INFERRED_RHYTHM` / `RESEARCH_SYNTHETIC`; app renders only the first two.
8. `RESEARCH_SYNTHETIC` is quarantined: watermarked, written only under `research/runs/`, no import path into `app/` or `inference/` (enforced by import-audit test).
9. "ECG" appears in user-facing text only in the referral sentence; no ECG-paper visual mimicry (grid style, lead labels, mV/mm-s notation are forbidden tokens under test).
10. No demographic features in any classifier; no non-permissive dependencies or RAIL-licensed checkpoints.

4. Workstreams

WS-A — Product core (prompt v0.2 M1 + v0.2.1)

Item	Detail
Heads registry	`heads/base.py`: `EndpointHead(name, version, required_inputs, run(BeatLattice, ctx) → HeadResult)`; config-flagged per head
`head_afib`	Wrap existing Model A + decision logic; behavior locked by regression tests (bit-identical sentence + stars on fixture corpus)
`head_rate_flags`	Sustained brady <50 / tachy >100 from clean-run rate; abstention rules identical to AFib head
`ScanResult v2`	Per-head results, measurement classes, capture metadata; schema version bump + v1 migration shim
Findings report	Clinician-style layout (header / measurements / findings / referral / footer), **findings-only default** (`report.mode`), printable single page, forbidden-content tests
CLI	`process --heads … [--report out.html]`

Acceptance: all v0.1 tests green; new snapshot + forbidden-content tests; demo shows the new report end-to-end incl. a NO_RESULT run.

WS-B — Data engine (v0.2 M2; the moat)

Context every engineer should know: **no public dataset pairs facial video with ECG during arrhythmia** (verified Aug 2026). The clinical dataset does not exist until we build it, and the field's weakest spots are exactly what we will enroll for: hard negatives (PAC/PVC, flutter, sinus arrhythmia, motion) and Fitzpatrick V–VI skin tones (published no-read rates hit 39% in the darkest-complexion group; we treat per-tone coverage as a release gate, not a footnote).

Item	Detail
Campaign manifests	YAML: target N, quotas (Fitzpatrick I–VI, device matrix, lighting), consent version, per-session checklist; `cli.py campaign status`
Capture profile	60 fps target (30 min), per-frame hardware timestamps, AE/AWB locked where OS allows, pre-encode ROI processing; no compressed third-party video for rhythm work
Paired-ECG ingestion	`cli.py ingest-reference`: CSV/WFDB reference ECG; PRBS/LED sync verification; stored clock offset + drift; R-peak series; **video↔ECG alignment error <10 ms or fail closed**
Label schema	**Episode-locked** rhythm labels (label = time span on the ECG, never a patient attribute), adjudicator fields, ESC-definition checkbox; morphology fields (conduction pattern, measured PR/QRS/QT per segment) for the L8 track
Registry & splits	`datasets/registry.jsonl` (content hashes, lineage, consent class); splits derived from registry: participant + session disjoint; leave-one-device/site-out for validation runs

Dataset targets (from the feasibility review; ranges, to be refined against pilot yields): signal feasibility 30–60 participants; algorithm development 300–1,000; **≥150–300 ECG-confirmed AF-during-scan episodes; 300–600 hard negatives**; pivotal (locked model) 500–1,500; external validation 300–800 at ≥2 new sites.

WS-C — Training engine (v0.2 M3)

Item	Detail
Runs	Config-driven (`configs/train_*.yaml`), deterministic seeds; run record = dataset ids + split id + config hash + git rev + metrics
Mandatory baselines	tachogram-statistics, HR-only, demographic-only, **participant-history** (memorization detector), **device/site** (leakage detector) — implemented as heads-compatible models
Promotion	`cli.py promote` refuses unless: beats all baselines on disjoint splits, leakage audit clean, no-read parity within threshold per Fitzpatrick group; promotion appends to spec changelog
Model registry	`models/registry.jsonl` : semver, run record, eval report hash — the PCCP-shaped audit trail FDA will eventually want

Evaluation metrics (standing): Se / Sp / AUROC / PPV / NPV / F1, **false positives per 1,000 scans**, **no-read rate** — all stratified by skin tone, device, age, sex, lighting, and reported at deployment prevalence (PPV-vs-prevalence curves, not enriched-cohort PPV).

WS-D — ECG-reconstruction research track (v0.3; quarantined)

Purpose: build the full video→ECG estimation capability the long-term ambition calls for, **behind pre-registered promotion gates**, so the question "can reconstruction reach clinical fidelity?" is answered by our data rather than assumed either way. Components: `research/ecg_reconstruction/decoder.py` (permissive-license seq model; pretrain on cached MIMIC PERform PPG+ECG pairs; fine-tune on registered facial data), `fidelity.py` (gate evaluator), `cli.py reconstruct` / `gate-status` .

§G promotion gates (all required, on participant+session-disjoint held-out data; thresholds `REQUIRES_CLINICAL_SIGNOFF`): **G1** beat the identity-template baseline on morphology; **G2** interval fidelity (QT MAE ≤20 ms and beats RR-only QT regression; PR ≤20 ms; QRS ≤15 ms) with CIs; **G3** blinded cardiologist abnormality preservation ≥80%/80%; **G4** no confabulation on held-out rhythm classes; **G5** rhythm detection from reconstructions non-inferior to the measured path. While any gate is red: research-only, watermarked, findings-only user surface. Engineering expectation, stated openly: G1–G4 are predicted not to pass (the morphological information is physically absent from facial video); the gates make that an empirical, auditable result either way — and the same promotion machinery is what ships AFib/flutter heads, which ride the measurable signal.

WS-E — Growth heads (research-flagged; no user output until promoted)

`head_flutter_suspicion` (sustained regular ~140–160 bpm + abnormally low dispersion + cross-scan integer-ratio rate steps), `head_irregularity` (rhythm-agnostic), `head_burden` (abstention-aware AF-suggestive fraction across a registered scan series). Known physics limits to encode in labeling: fixed-block slow flutter is a documented miss; ectopy is the primary AF false-positive source; a scan cannot detect an episode not occurring during it (serial/passive scanning is the clinical product shape).

5. Delivery plan

Dependencies: WS-A has none (start now). WS-B needs hardware selection (reference ECG device) + IRB/consent groundwork in parallel. WS-C needs B's registry. WS-D needs B's morphology labels for its gates (its MIMIC pretraining can start immediately). WS-E rides A + C.

Phase	Content	Exit criteria	Indicative size*
P1	WS-A complete	v0.2-M1 acceptance; demo on findings report	S–M
P2	WS-B engine + pilot capture (30–60 participants, multi-tone/device)	<10 ms sync verified in the field; registry live; pilot IBI MAE ≤30 ms at rest across Fitzpatrick I–VI; type-VI coverage ≥70%	M–L (calendar-gated by recruitment)
P3	WS-C + first retrained AFib model	Candidate beats all baselines subject-independent; promotion machinery exercised end-to-end	M
P4	WS-D lab + gate scoreboard; WS-E stubs	`gate-status` runs on real paired data; falsification write-up in `docs/`	M
P5	Hard-negative campaign + locked evaluation (feeds the pivotal study design)	Pre-registered confusion-matrix report: AF vs PAC/PVC/flutter/sinus-arrhythmia/motion on unseen subjects & devices	L (clinical partnerships)

*S/M/L relative sizing on purpose — calendar estimates depend on team size and, for P2/P5, on clinical recruitment, which history says is the long pole. The 90-day research plan in the feasibility review maps to P1–P4 on an aggressive single-team schedule.

Suggested team shape: 1 signal/CV engineer (L0–L3), 1 ML engineer (L4/L7/L8), 1 data/backend engineer (L6, registries, pipelines), 1 capture/mobile-web engineer (demo, campaign tooling), a clinical data coordinator for campaigns, plus fractional clinical advisor (gate thresholds, adjudication) and regulatory consultant (Q-Sub prep). QA is everyone: tests-first is the repo's law.

6. Risk register (top items)

Risk	Impact	Mitigation
Video↔ECG sync worse than 10 ms in field conditions	Corrupts all beat-timing labels	PRBS/LED marker per session; fail-closed ingest; drift checks; P2 exit criterion
Darker-skin no-read / accuracy gap (field-documented up to 39% no-read)	Clinical, equity, and regulatory failure	Multi-ROI SNR weighting; ≥100 lux guidance; per-tone quotas + coverage as release gate; fairness telemetry in every eval
Ectopy/flutter false positives (field's known weak spot: 61.8% other-arrhythmia accuracy in best LOSO study)	Unshippable FP burden	Hard-negative campaign (P5); confusion-matrix exit criteria; flutter labeled as limitation
Shortcut learning (identity/device/site)	Inflated metrics that collapse externally	Mandatory baselines incl. participant-history + device/site; leakage audit in every eval
Gate erosion under demo pressure	Ships fabricated waveforms; existential credibility risk	§G thresholds changeable only via signed spec changelog; forbidden-content + import-audit tests in CI
Recruitment slower than plan	P2/P5 slip	Parallel site partnerships; passive-capture option; campaign dashboard early
Regulatory boundary drift in copy/UX	Wellness→device reclassification exposure	"ECG"-token tests; sanctioned-sentence gate; regulatory review of all user-facing strings

7. Where everything lives

Execution prompts (run with Claude Code, in order as needed): `CLAUDE_CODE_PROMPT_v0.2.md` (platform: heads/data/training), `CLAUDE_CODE_PROMPT_v0.2.1_clinician_report.md` (+ v0.2.2 findings-only addendum), `CLAUDE_CODE_PROMPT_v0.3_ecg_reconstruction_gated.md` (reconstruction lab + §G). Authority: `CLAUDE_CODE_SPEC.md` (append-only changelog). Evidence base: `AvatarX/research/AvatarX_Contactless_ECG_FaceScan_Feasibility.pdf` (50-page review: physiology, evidence tables, validation program, FDA landscape) and the 2-page plain-English brief. Questions of "why is X forbidden/gated" resolve to the review's §12 (reconstruction verdict), §17 (no-read architecture), §18 (leakage/baselines), and §22 (regulatory).

Prepared from the August 2026 feasibility investigation. Accuracy figures referenced here are controlled-study ceilings, not deployment expectations; nothing in this plan constitutes a clinical claim. Numeric gate thresholds marked for clinical sign-off are provisional.